

ALPINE DROUGHT OBSERVATORY

Report on existing monitoring platforms and
potential data for the integration into ADO
(D.T1.1.2)

Prepared by Slovenian Environment Agency (ARSO)
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ADO*

Alpine Drought Observatory (ASP940)

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1. INTRODUCTION

Purpose and scope of the report on existing monitoring platforms and potential data for the integration into ADO (Deliverable D.T1.1.2)

The ADO project aims at developing a platform for improved and harmonised drought monitoring for broader Alpine area (the Alps and their surrounding lowlands as defined by the Alpine Space programme; most of the area is covered by six countries: Austria, France, Germany, Italy, Slovenia and Switzerland). Monitoring different aspects of the drought cycle in Alpine area may require a variety of indicators and indices, which will depict the drought conditions on the very diverse and inhomogeneous terrain of Alpine region. Given the importance of the Alpine region for local and downstream water-related activities, there is a commitment of ADO project to provide information on different characteristics of the hydrological and meteorological variables triggering drought situation in the Alpine region and to display different stages and types of the drought development.

The purpose of this report is to prepare a foundation for this working process. An overview of the state of drought observation and detection in the Alpine region has been prepared, which will also serve as basis for its upgrade or extension to the entire Alpine area. The state of the art analysis prepared by project partners (at least one per participating country) was conducted by uniform **questionnaire** that was circulated among ADO project partners covering four topics: drought monitoring approach, drought-information platforms, the regional & EU drought monitoring approach, and potential drought indices in ADO.

The report aims to serve as a starting point, showing the main indicators/indices which are already available and being put into practice in countries' regions covered by the project. However, this report does not intend to recommend a final short list of indicators and indices; rather it presents a broader set of indices and indicators, that have already been applied in the region or by competent institutions outside of regions, or have been recommended by project partners (through questionnaire) or found in relevant literature. The final shortlist will be based on specific characteristics of drought most closely associated with the input data, their availability, applicability and impacts.

Through filled-out questionnaire, which addresses an overview of existing drought monitoring in the Alps and capitalize existing products and knowledge, information were directly collected by all project partners through filling out the questionnaire, enriched with the help of key stakeholders and observers via interviews and publicly available data. Filled out questionnaires present a channelled review of the national drought monitoring's main characteristics (activity in thematic work package T1, deliverable D1.1.1).

A regional report (activity in thematic work package T1, deliverable D.T1.1.2) was carried out by Slovenian Environment Agency - ARSO (T1 leader). It was made on the basis of received filled-out questionnaires, a contribution by the ADO project partners. Since the gain of the information is still in the initial phase (also has been hindered due to pandemia) it will be enriched throughout the project development and more intense interactions with national stakeholders. Consequently, the scope of the assignment was in some countries limited and the review could focus only on a selected number of aspects. The review outlined below is considered to be a living document that will evolve and integrate new findings during the project development and input from target groups. Involvement of a wider audience will be facilitated through the activities, which are included in the communication work-package, i.e. stakeholder workshop, national project promotion, and within activity A.T1.1.

After collection of all national questionnaires (6), a **common regional review** of drought related data in participating countries is prepared with the 4 main objectives:

- (1) generate a first review of existing indices, indicators or drought monitoring approach for early detection of drought over the Alpine area, including drought-related data being collected for monitoring of different drought types (meteorological, agricultural and hydrological), early warning system and other information collected by vulnerable sectors;
- (2) prepare characteristics of available indices (methodology of calculation, input data and their availability, coverage etc) and collect information about the specific indicator from literature such as WMO's Handbook of drought indicators, guidelines of the Integrated Drought Management Program, the European Drought Observatory and the outcomes of DriDanube project or other drought-related projects.
- (3) collect information about existing drought monitoring platforms for end-users, available on country or regional level;
- (4) select or recommend indices that could be useful for Alpine-wide drought monitoring and consequently shown in the ADO platform, considering the specific requirements related to the diverse Alpine terrain.

The presented report aims represents the driver for additional studies/work within the ADO project.

To greatest extent it is connected to T2 work package, where collection and harmonization of data in Alpine space will be performed upon the findings of the first cross-section of drought indicators and methodologies already present in the report, especially of high value is the obtained first list of potential indices or indicators that could be applied throughout the region.

Additionally, set of indicators will build the potential for the research to what extent they reflect drought impacts, specific to the Alpine Space (T3). Revision of considered indicators/indices will be checked through the feedback from case study regions (working package T5) and in the process of ADO platform creation (working package T4).

2. DROUGHT INDICATORS IN USE

The indicators and indices (hereinafter: indices) in use in Alpine parts of the project countries are presented in the Annex table (*Collection of existing and proposed drought-related indicators and indices for potential integration into ADO platform*).

Comment on the overview table (Annex):

- The table includes indices and information that were extracted from the channelled questionnaires, filled out by project partnership. It does not present the entire set of indices in use in the 6 Alpine countries, nor are they all operatively in use at national level. Several of them are used locally or just for the Alpine part of the country although for the sake of simplicity, the indices are structured as per country names. Considering all the mentioned limitations of the gathered information, the table should be used experimentally.
- The content of the table relies on the wide scope of internal expertise and knowledge of the ADO partnership, thus this set of indices is considered representative enough to carry out further discussion and work in developing ADO platform. Therefore, the Annex table presents the basis for simple analysis done in this chapter, and the base for further discussion between thematic work packages of ADO project throughout project lifetime.
- Selection of indices to be proposed for further discussion and work on them was carried out through several online meetings between project partners, mainly those involved in T1 and T2 activities, hydrological experts from T3, and T5 leaders from pilot-case and drought-management point of view. First step in shortlisting the indices presented marking those who have theoretical potential to be applied in the same way across to all Alpine area, in other words to exclude those that require specific soil or plant characteristics. First-step-selected indices are marked in column "Proposed for further discussion" of the Annex file.

Findings from the overview table (Annex):

- There are altogether 34 different indices operationally in use in Alpine area to monitor the state of drought. In addition to those, other indices were mentioned in the questionnaires that primarily refer to detection of ecological status of waters, under-development indices or methods that can be applied to different datasets. Due to the nature of the latter, which were different from the rest, they were not included in simple analysis done in this chapter.
- Indices written in grey likely address the same field of interest, but only use different approach for its calculation. Thus are counted as one index. However, since such indices are calculated through a different approach, depending on a country, it is not to be directly assumed that the outcomes of same index calculation are comparable among the regions.

2.1 OVERVIEW OF INDICES USED BY COUNTRY AND BY DROUGHT TYPE ADDRESSED

The indices in use describe changes in parameters that can be related to four different types of drought:

- **Meteorological drought:** indices describe changes in atmospheric parameters,
- **Agricultural drought:** some indices describe changes in topsoil parameters, while others describe changes in vegetation health parameters,
- **Hydrological drought:** some indices describe changes in surface water parameters, while others describe changes in groundwater parameters,

The two pie charts below present the share of all drought indices used in two ways: as per country use, and as per drought type addressed. Figure 1 shows number of indices used in each region for monitoring different types of drought. Figure 2 shows number of indices used in all Alpine area for monitoring each of the drought type. Altogether, the split number of indices used for drought monitoring as per each Alpine country and as per the type of drought they address are presented in Table 1 further below.

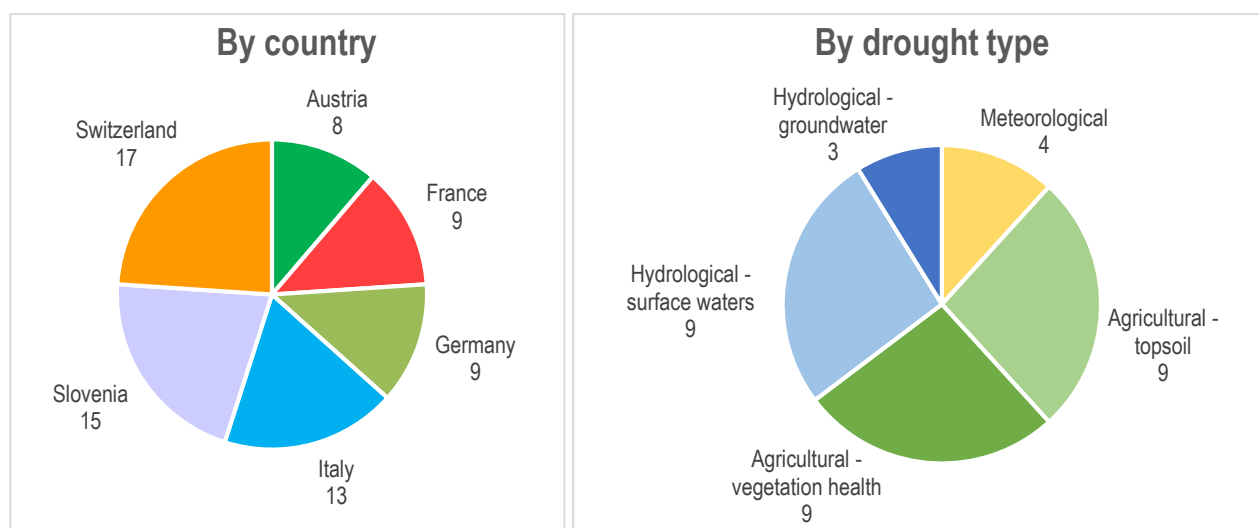


Figure 1 (left): number of indices used in each region for monitoring different types of drought. Figure 2 (right): number of indices used in all Alpine area for monitoring each of the drought type.

Table 1: the split number of indices used for drought monitoring in each Alpine country as per the type of drought they address.

Used in country per drought type addressed	Meteorological	Agricultural		Hydrological		All
		– topsoil	– vegetation health	– surface waters	– groundwater	
Austria	2	1	2	2	1	8
France	3	1	1	3	1	9
Germany	4	1	1	2	1	9
Italy	3	3	1	4	2	13
Slovenia	3	5	2	3	2	15
Switzerland	3	5	2	6	1	17

Findings from the overview:

- A variety of indices are used to monitor drought conditions, and all countries have at least one index operationally in place for each of the three main components of hydrological cycle – atmosphere, surface layer and groundwater. In addition to that, two countries also operatively monitor parameters that address ecological status of waters, which may worsen in case of hydrological drought.
- There is a higher number of indices used to monitor agricultural drought and hydrological drought of surface waters, which may be a result of complexity of the system they address – soil, vegetation, snow, rivers and lakes.
- Lower number of indices used to monitor meteorological drought and hydrological drought of groundwater may be a result of lesser parameters involved in describing the starting- and end point of water flow in hydrological cycle, respectively – atmosphere, groundwater.

2.2 OVERVIEW OF INDICES BY TYPE OF CALCULATION

In terms of the index calculation approach, values of most of the indices operatively in use in Alpine countries are derived directly from measurements data from meteorological or hydrological gauge station (*i.e. share of precipitation, Q_{min} , groundwater level*), or they are calculated through simple equations (*i.e. drought magnitude, surface water balance, streamflow level in percentiles*). A great deal of operational indices is derived also using specific models to simulate addressed conditions (*i.e. Evaporative Demand Drought Index, Plant Water Deficit or Snow Water Equivalent*). Only five of the indices are obtained using remote sensing approach, through radar data and satellite data (*i.e. Soil moisture in Italy, fAPAR or NDVI*).

The share of indices operationally in use by the type of their calculation is presented in Figure 3 below.

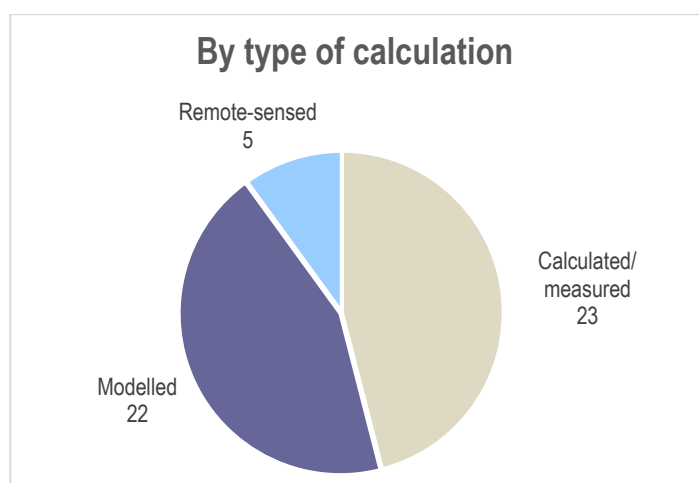


Figure 3: the share of indices operationally in use by the type of their calculation.

Findings from the overview:

- An overlapping of type of calculation can be observed in certain indices. The reason for this lies in the fact that some indices are derived through multi-step approach, meaning some steps are considered calculated while further steps require modelling, and some other indices are obtained using data derived from different sources, *i.e. measured, modelled and remote-sensed*. Therefore, such indices were included in more than one category of calculation type of calculation category. This can be more thoroughly reviewed in Annex table.

2.3 OVERVIEW OF INDICES BY PROJECTION TYPE

In terms of spatial projection of index value, most of the indices operatively used in the Alpine countries are displayed as a point value, less so are gridded in a net-display while only seven of them are displayed as interpolated.

The share of indices in use by their projection type is shown in Figure 4 below.

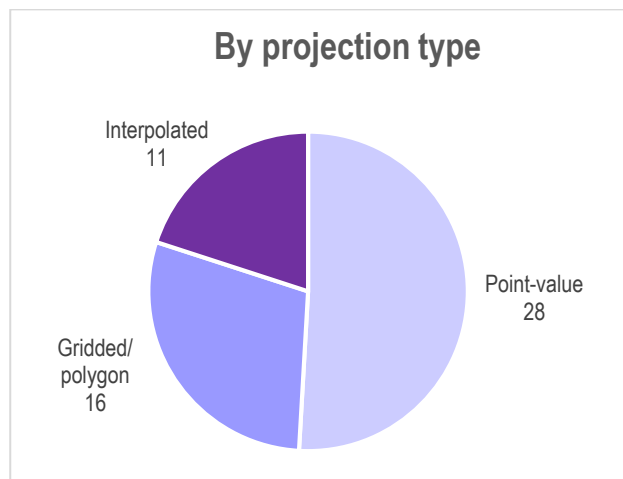


Figure 4: the share of indices in use by their projection type.

Findings from the overview:

- Point-value information is the base for the projection of most of the indices. However, quite some of the indices are projected (or allow projection) in multiple ways, i.e. as point-value and interpolated, thus the overlapping in projection type for some indices. This can be more thoroughly reviewed in Annex table.
- It should be noted that certain indices take into calculation the characteristics of the exact point they are calculated for, and thus do not allow further projection as gridded or interpolated. These mainly include agricultural drought indices that take into consideration topsoil texture characteristics or plant-specific characteristics, and hydrological drought indices taking into account streamflow data or groundwater levels.

3. EARLY WARNING SYSTEM IN THE ALPINE AREA

3.1 GENERAL OVERVIEW OF EARLY WARNING SYSTEM APPROACH

A drought early warning system's main purpose is to warn local communities upon the risk of drought, improving preparedness and decreasing risk associated with crop, food and other losses. Early warning system (EWS) requires drought monitoring using appropriate drought indicators, meteorological data and forecast; a warning signal; public awareness and education, institutional cooperation and data sharing (*Source: Climate change adaptation technologies for water: a practitioner's guide to adaptation technologies for increased water sector resilience, UNEP-DHI Partnership, UNEP-DTU, CTCN, 2017*)

This chapter focuses how drought severity and a warning signal is displayed in the Alpine area/Alpine countries and touches some other topics related to EWS approach, which appeared relevant for a display.

In T1 questionnaires, project partners in cooperation with project observers and other key stakeholders (if needed) obtained information about early warning system approach in their country for the Alpine area on the base of following questions and attached links to the relevant websites:

- How is drought severity/warning level displayed?
- What indices, data and information build each of the warning levels?
- Is it displayed regularly (up-to-date status) or when needed?

The results of the questionnaire's feedback are collected in Table 2 below.

Table 2: Analysis of EWS approach with focus on the level of EWS formality in the country, its organisation in general and display of drought severity.

Country	Level of EWS formality		Relation to measures	Organisation of EWS approach			Display of drought severity		
	Official	Not official		Country level	Regional	River basin	Maps Severity classes	Colour scale not classified	Bulletins/info on webpage
Austria		(✓)		✓			✓ (forest fire risk index)	(✓)	(✓)
France	✓		✓	✓		✓	✓		✓
Germany		✓		✓			✓	✓ (aridity index)	✓
Italy		✓				✓ Po		✓	✓
		✓			Emilia Romagna			✓	✓
		✓			Piemonte				✓
		(✓)			Other regions				(✓)
Slovenia		✓		✓					✓
Switzerland		✓		✓			✓	✓	✓

(✓) not direct information on drought occurrence in the country or region

The analysis of the feedback from the questionnaires shows the following:

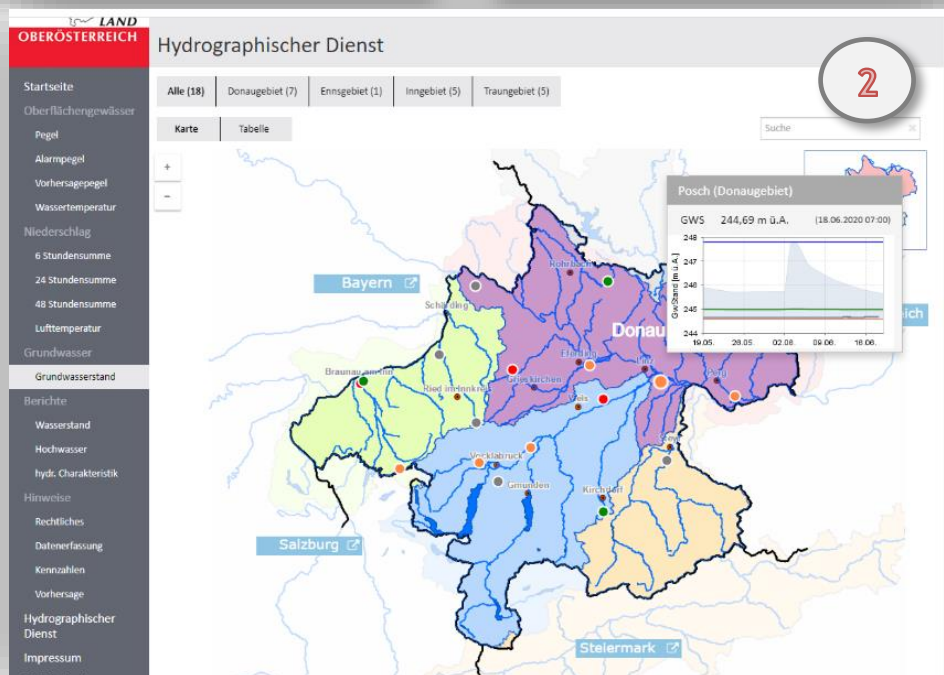
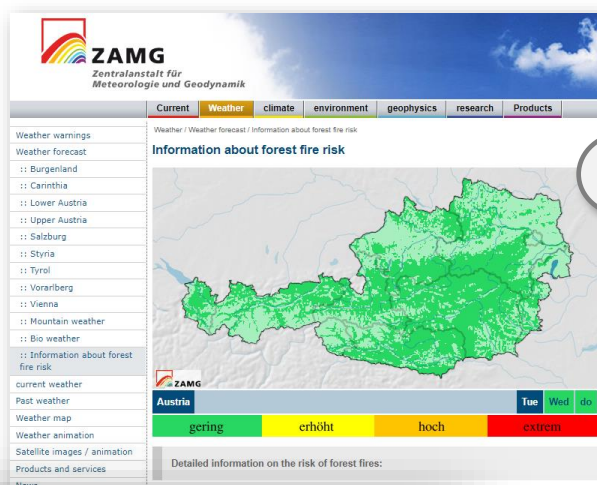
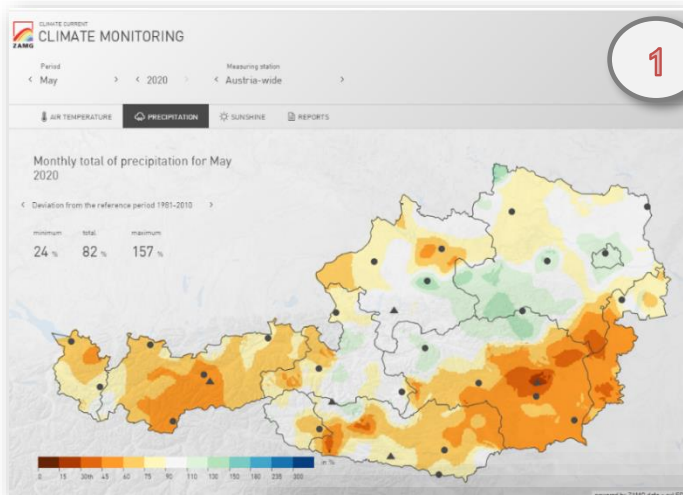
- EWS approach vary greatly from country to country in Alpine area. Differences are related to the level of EWS formality in each country, its organisation in general (country/regional level), number and type of used drought indices and a way of communication or display of drought severity to the public (classified severity levels or not; information given via platform, maps or/and bulletins).
- In most countries, EWS is in general organised at country level and further adapted to regional/local specifics, while in others, i.e. Italy, drought monitoring and EWS approach differs from region to region.
- In only one country, France, there is an official, legally binding EWS with defined drought indices and a scale for drought levels which are related to specific measures. In other countries, EWS is not legally binding, meaning the experts assess drought severity on the base of their operational drought indices, provide information to public, although drought severity levels are not related to specific measures for prevention or mitigation of damage caused by drought.
- There is a difference in a display of drought severity: in most cases, drought indicators are classified in severity levels (Switzerland, France, Germany, for most of the indices) while in some other regions, values of drought indicators are displayed through a colour scale (in case of a region in Italy) but not divided into drought severity classes.
- Information to public about drought is usually displayed via webpages of institutions responsible for monitoring drought and research institutions, or via platforms designed especially for this purpose (France, Switzerland, Germany-Bavaria, Italy-Emilia Romagna) in a form of maps usually with links to data, graphs, or/and bulletins and information on the webpage. In case of Austria, there is no direct information on drought occurrence, however there is display of deviation of temperature and precipitation from the long-term average on the country level with accompanied text on the webpage text, and also display of hydrological parameters compared with relative statistic values on the regional level. In Slovenia, the warning about drought and its severity is currently provided to public via Drought bulletin and information to different media. On a map there are displayed just reported impacts in agriculture.

For detailed information per country, see description for respective country below.

3.2 DETAILED INFORMATION ON EWS IN ALPINE COUNTRIES

• EWS approach in Austria

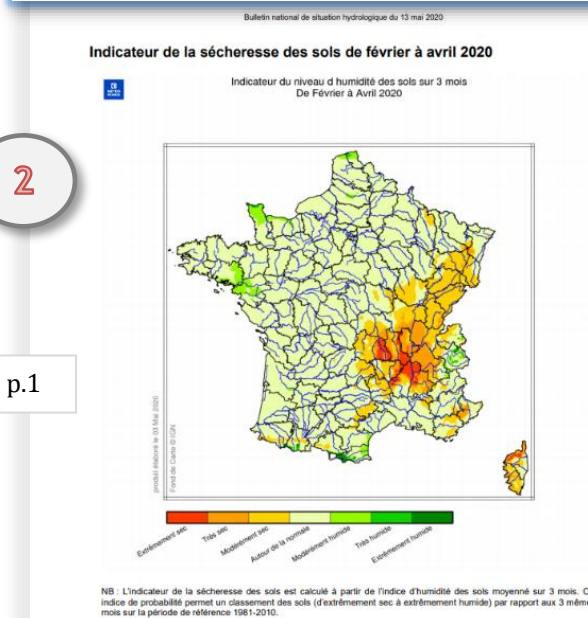
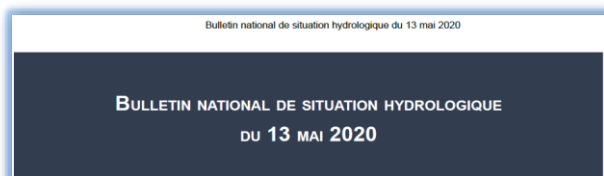
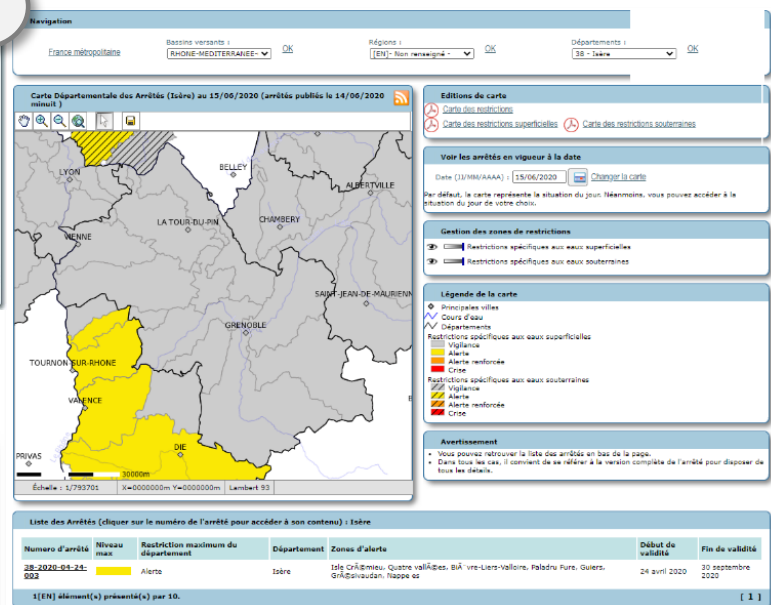
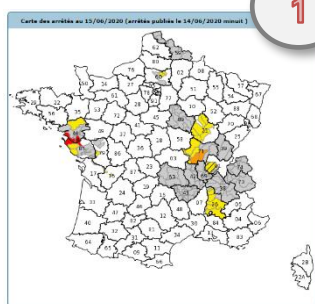
Level of formality	Display of drought severity >> Communication to public	Link:
In Austria there is no early warning system referring to drought severity via drought indicators and related communication to public. However, on the base of monitoring meteorological and hydrological parameters, there is a display of temperature and precipitation deviation from the long-term average on the country level and on the regional level monitored hydrological parameters compared with relative statistic values. Additionally, for the Austria forest fire risk index is calculated and classified to warning levels which are related to measures.	Country level: Maps - Climate monitoring of <u>air temperature and precipitation</u>; deviation compared to reference period 1981-2010 (links to graphs) Weather review/reports	1. https://www.zamg.ac.at/cms/de/klima/klima-aktuell/klimamonitoring/?view=fullscreen&param=t&period=period-y-2020&ref=3 Prepared by ZAMG
	Regional level: e.g. Upper Austria Maps - <u>Hydrological status</u> of surface water and groundwater compared with relative statistic values Links to graphs and data	2. http://hydro.ooe.gv.at/ Prepared by Office of the Government of Upper Austria, Hydrographical Service Upper Austria
	Country level Maps - information on current <u>forest fire risk</u>, which is calculated from meteorological parameters (air temperature, atmospheric humidity, wind velocity, precipitation) and quantified in four hazard levels.	3. https://www.zamg.ac.at/cms/de/wetter/wetter-oesterreich/waldbrand Prepared by ZAMG



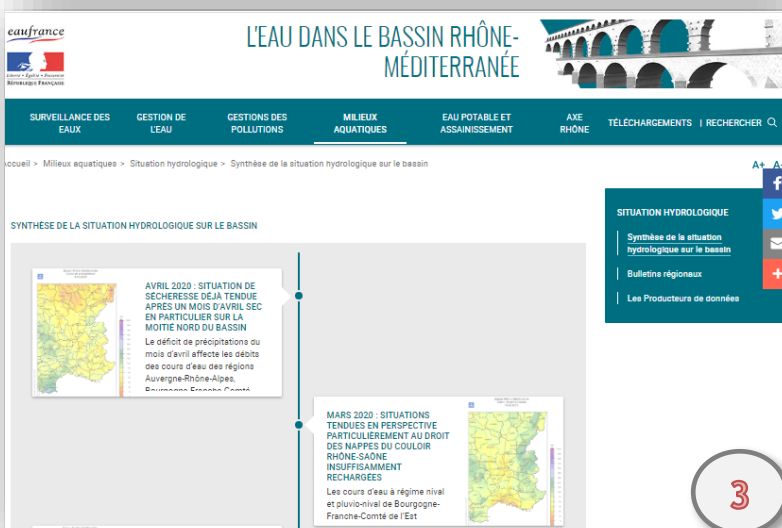
• EWS approach in France

Level of formality	Display of drought severity >> Communication to public	Links:
Official/formal EWS protocol on national level with defined drought indices and scale for drought levels (for all types of drought). Thresholds for drought indices are specified at local levels (NUTS3). Drought levels are related to specific measures (river flows, piezometric levels on reference points).	Country and NUT3 level: Up-to-date maps - official decrees for water use restriction	1.Platform: http://propluvia.developpement-durable.gouv.fr/propluvia/faces/index.jsp Prepared by Propluvia; Ministère de la Transition écologique et solidaire; Ministère de l'Agriculture et de l'Alimentation
	Country level: Monthly bulletins on hydrological situation (included precipitation, soil, surface and groundwater, impacts on aquatic ecosystems)	2. https://www.eaufrance.fr/publications/bsh Prepared by eaufrance, Le service public d'information sur l'eau
	Basin Rhone-Mediterranean: Monthly bulletins on hydrological situation (included precipitation, soil, surface and groundwater, impacts on aquatic ecosystems)	3. https://rhone-mediterranee.eaufrance.fr/milieux-aquatiques/situation-hydrologique/synthese-de-la-situation-hydrologique-sur-le-bassin Prepared by eaufrance, Agence de l'eau Rhône-Méditerranée

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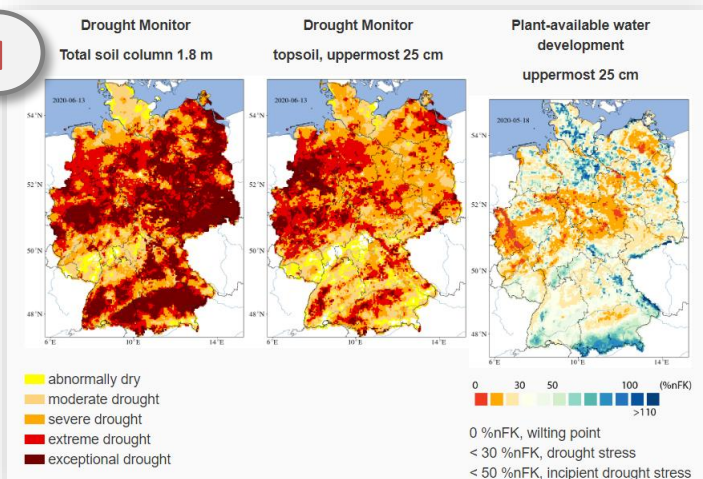


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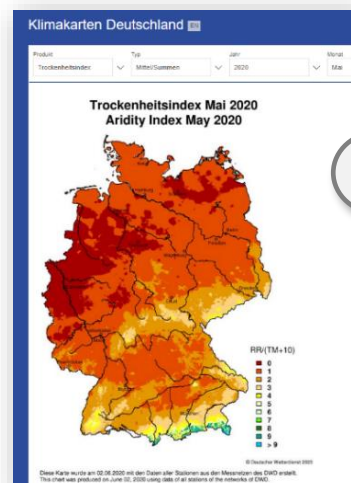
• EWS approach in Germany

Level of formality	Display of drought severity >> Communication to public	Link:
Expert assessment on the base of different drought indices. None has official, legally binding character for decisions. Drought severity levels are not related to specific measures.	Country level: Maps - daily information on agricultural drought; classified in drought levels (information is based on simulations using the mHM meso-scale hydrological model developed at the UFZ) Maps could be downloaded	1. UFZ Drought Monitor website: https://www.ufz.de/index.php?en=37937 Prepared by UFZ Helmholtz, Centre for Environmental research
	Country level: Maps - Aridity index monthly; not classified in drought levels	2. https://www.dwd.de/DE/leistungen/klimakartendeutschland/klimakartendeutschland.html?nn=18256 Prepared by DWD (Deutscher Wetterdienst/German weather service)
	Bavaria: Maps - Daily Information on meteorological drought, low flows and low groundwater level, (entire Bavaria and sub basins) classified in drought levels/classes of humidity; in case of ground water: no low/low/very low/new lowest value + links do graphs, data + text information on the website	3. Platform Low water information service Bavaria https://www.nid.bayern.de/ Prepared by Bavarian State Office for the Environment

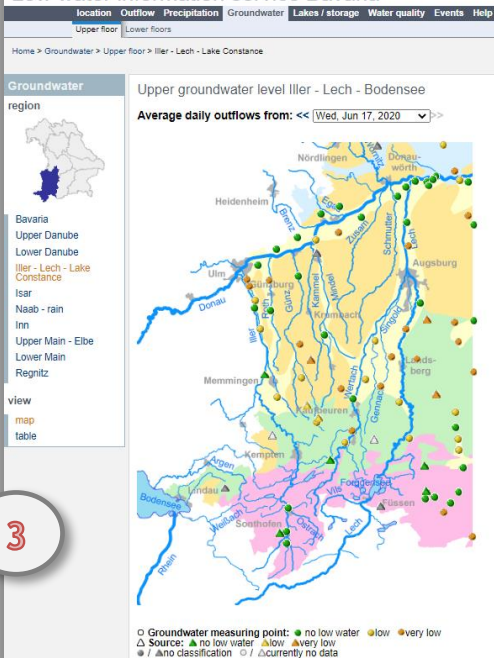
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2



Low water information service Bavaria

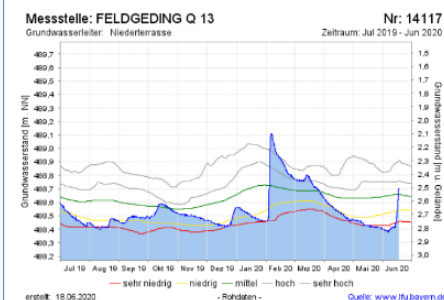


3

FELDGEDING station Q 13

Groundwater levels in the past 12 months

Situation: **no low water**
 groundwater level [m u. NN]: **488.69**
 groundwater level below site [m]: **2.52** site
 height [m u. NN]: **491.21**
 Last measurement from June 17, 2020



Groundwater levels from: Wednesday, June 17, 2020

Measuring point	Lkr.	Aquifer	date	Groundwater level [m u. NN]	Groundwater level [m u. Terrain]	situation
Garmisch-Burgrain D / 1K	CAP	quaternary	June 17, 2020	673.71	2.57	very low
PENZING II FB 824	LL	quaternary	June 17, 2020	598.24	8.85	very low
SEND 68	NU	quaternary	June 17, 2020	482.98	3.57	very low
more ...						

Source fillings from: Wednesday, June 17, 2020

Measuring point	Lkr.	Aquifer	date	Spring fill [l / s]	situation
Bischofsried	LL	Gravel areas	June 17, 2020	76.24	very low
Gold gutter Oberrieden	MN	Upper fresh water molasse	June 17, 2020	2.27	very low
Steindorf dam	AJC	moraine	June 17, 2020	0.76	very low
more ...					

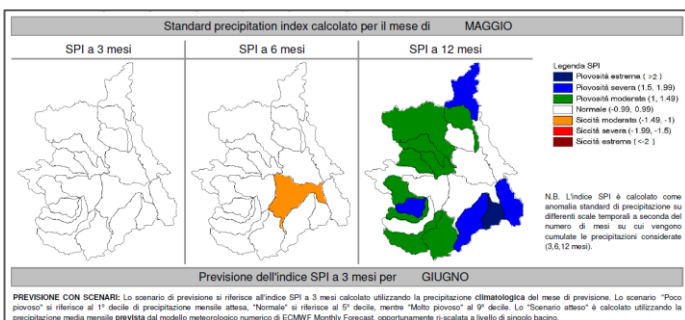
• EWS approach in Italy

Level of formality	Display of drought severity >> Communication to public	Link:
In Italy EWS approach varies between institutions and/or regions. There is no standard methodology to define a drought warning level and status of drought severity is usually based on an expert assessment made on the base of different drought indicators status. Drought severity levels are not related to specific measures. In Italy, weather and climatic services are usually provided by the »Agenzie Regionale per la Protezione dell'Ambiente« (ARPA), which are independent environmental agencies for each Italian region. Only in some regions as Piemonte and Emilia-Romagna bulletins or maps with drought indicators are regularly published.	<i>Piemonte region</i> Monthly bulletin on hydrological situation; Weekly hydrological bulletin in late spring and summer to update the monthly hydrological bulletin (in case of drought)	1. http://www.arpa.piemonte.it/bollettini/elen-co-bollettini-1/bollettino-idrologico Prepared by ARPA
	<i>Region Emilia-Romagna:</i> Maps – many different drought indicators (values not classified in drought severity levels; colour scale) and water availability (below/above the average) >> presented also in Monthly bulletins with links to maps, graphs; Local bulletins; Crisis bulletins	2. https://ambiente.regione.emilia-romagna.it/it/acque/temi/siccita Prepared by: Servizio Tutela e Risanamento Acqua, Aria e Agenti Fisici
	<i>Po River district:</i> Monthly bulletins on hydroclimatic situation	3. https://adbpo.gov.it/osservatorio-permanente/ Prepared by: Po River District Authority
	<i>Other regions: Veneto, Friuli Venezia Giulia, Lombardia, Liguria, Trentino-Alto</i> Drought is only reported if exceptional situations occur and only in hindsight (in monthly reports on the climatic situation) but no outlook or assessment of the current situation. Arpa Liguria publishes an SPI map with their monthly climate reports.	

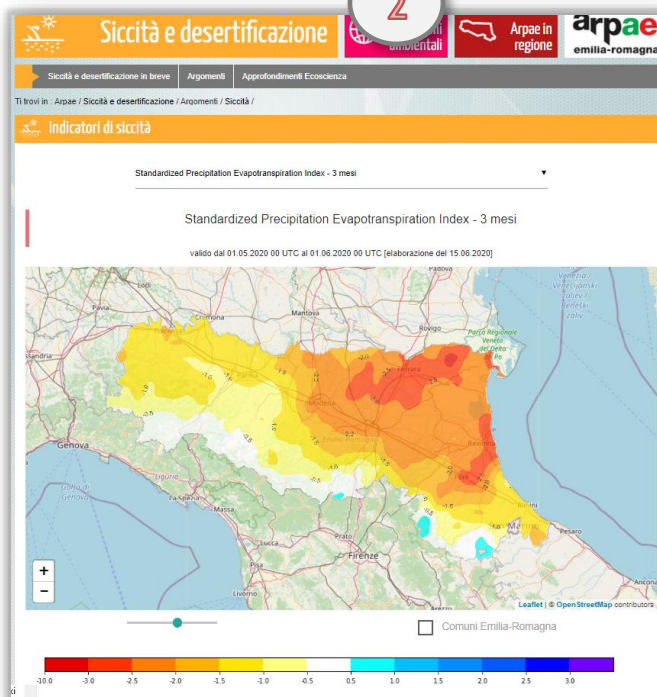
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BOLLLETTINO N°	DATA EMISSIONE	VALIDITA'	AGGIORNAMENTO	SERVIZIO A CURA DI	AMBITO TERRITORIALE
05/2020 Pag. 1/4	04/06/2020	MAGGIO 2020	MENSILE	ARPA - Rischi Naturali e Ambientali	Regione Piemonte



2



3



- **EWS approach in Slovenia**

Level of formality	Display of drought severity >> Communication to public	Link:
Drought warning is based on an expert assessment of the current status of different drought indicators. Drought severity levels are not related to specific measures yet.	<i>Country level:</i> Monthly bulletins - information on meteorological, agrometeorological and hydrological conditions in Slovenia	1. https://www.arso.gov.si/o%20agenciji/knji%c5%be nica/mese%c4%8dni%20bilten/ Prepared by ARSO
In the frame of national reporting network (established within DriDanube project), state of soil and vegetation/crops are weekly reported by on-field reporters.	<i>Country level:</i> In case of drought weekly Drought bulletins	2. http://meteo.arso.gov.si/met/sl/agromet/drought/ Prepared by ARSO
	<i>Country level:</i> Maps - estimated drought impacts on main crop yield	3. http://meteo.arso.gov.si/uploads/probase/www/agromet/product/form/sl/dridanube/karte_NRN.html Platform: https://droughtwatch.eu/

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- Hidrometeorološke razmere 4.6.2020 (pdf)
- Hidrometeorološke razmere 21.5.2020 (pdf)
- Hidrometeorološke razmere 14.5.2020 (pdf)
- Hidrometeorološke razmere 4.5.2020 (pdf)
- Hidrometeorološke razmere 23.4.2020 (pdf)
- Hidrometeorološke razmere 16.4.2020 (pdf)

Spremljanje aktualnih sušnih razmer je omogočeno na portalu [droughtwatch.eu](#), razvitim v okviru projekta DriDanube.

Ocenjen vpliv suše na pridelek kmetijskih rastlin

Karta opažanj s terena prikazuje tedenske ocene vplivov suše na pridelke v zgornjem sloju tal s strani poročevalcev. Poročila ugotovitve pa sporočijo preko spletnega vprašalnika [questio](#) statističnih regij Slovenije. V poročevalski mreži lahko sodeluje vsak zainteresirani posebej prihodek kmetijskih kultur preko [spletnega vprašalnika](#).

Ocenjen vpliv suše na pridelek kmetijskih rastlin, pridobljen iz poročevalske mreže DrDanube

3

Zbirni teden: 2020/22 Obdobje: 25.05.2020 do 31.05.2020

1. OCENA VPLIVA SUŠE NA PRIDELEK KMETIJSKIH KULTUR

Podatki so zbrani za obdobje 2020/22 (25.05. - 31.05.2020)

2. VODNA BILANCA ZA OBLASTI DRDANUBE

2. VODNA BILANCA ZA OBLASTI DRDANUBE

3. VLAŽNOST V ZGORNJEM SLOJU TAL

3. VLAŽNOST V ZGORNJEM SLOJU TAL

4. VLAŽNOST V ZGORNJEM SLOJU TAL

4. VLAŽNOST V ZGORNJEM SLOJU TAL

1. Oštevilčena ravnina

- 1. Suša brez vpliva na pridelek
- 2. Suša z varjetnim vplivom na pridelek
- 3. Suša z odličnim zmernjenjem pridelek
- 4. Suša z močnim zmernjenjem pridelek

2. Oštevilčena ravnina

- 1. Suša brez vpliva na pridelek
- 2. Suša, pridelek bi nekoliko zmernjen
- 3. Suša, pridelek bi močno zmernjen

3. Ekstremno sušne razmere - močni negativni vplivi

- 1. Zelo huda suša - opazni negativni vplivi
- 2. Zmerno sušne razmere - negativni vplivi
- 3. Sušna suša - zmerno sušne razmere - negativni vplivi
- 4. Zelo vlažne razmere - opazni negativni vplivi
- 5. Ekstremno vlažne razmere - močni negativni vplivi

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Hidrometeorološke razmere in stanje vodnih zalog v Sloveniji

04. maj 2020

Meteorološke razmere

2

V večjem delu Slovenije so se sušne razmere rahlo izboljšale, razen na Obali in delu vzhodne in severovzhodne Slovenije, kjer se sušne razmere nadaljujejo.

Prva četrtina leta 2020 je izjemno sušna. Od začetka marca pa do zadnjega dela aprila nikjer v Sloveniji ni bila zabeležena pomembnejša količina padavin. Bilo je sicer nekaj prehodov hladnih front, ki pa so žal povzročile le težave zaradi nizkih temperatur zraka; padavin pa je padlo le nekaj mm, na zahodu Slovenije (Obala, severna Primorska) v mesecu in pol ni padlo skoraj nič padavin. Skupna količina padavin od začetka leta do prvih dni maja izkazuje precejšen primanjkljaj, saj je padla le polovica povprečnih padavin.

Padavinski dnevi ob koncu aprila in v prvih dneh maja so prinesli največ padavin notranjski, kočevski, zgornjesavski in bovški regiji, skupno od 35 do 72 mm padavin. V osrednjem delu Slovenije, od Gorenjske pa vse do Bele krajine je deževalo v večjih padavinskih intervalih, ki so prinesli od 25 do 35 mm. Skromnejša pa je bila padavinska slika v vzhodnem delu Slovenije (do 25 mm), najslabša pa je padavinska situacija na Obali, kjer je, razen posameznih ploh, ostajalo suho.

Portoraj - kumulativna količina padavin

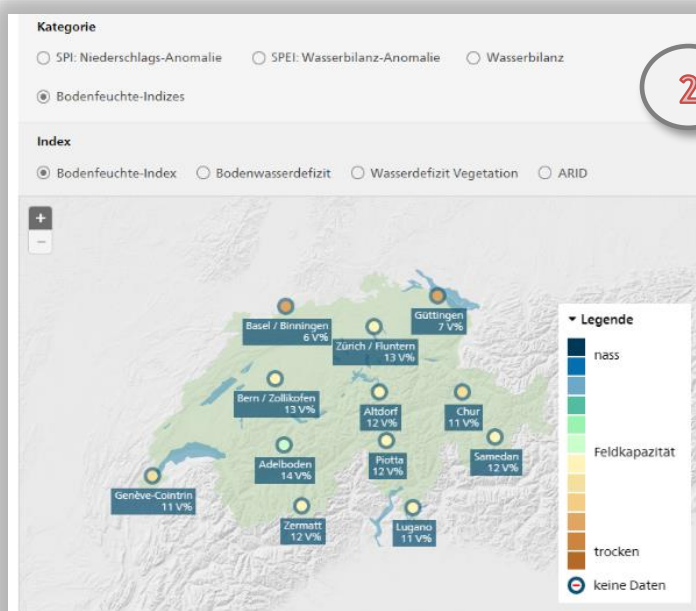
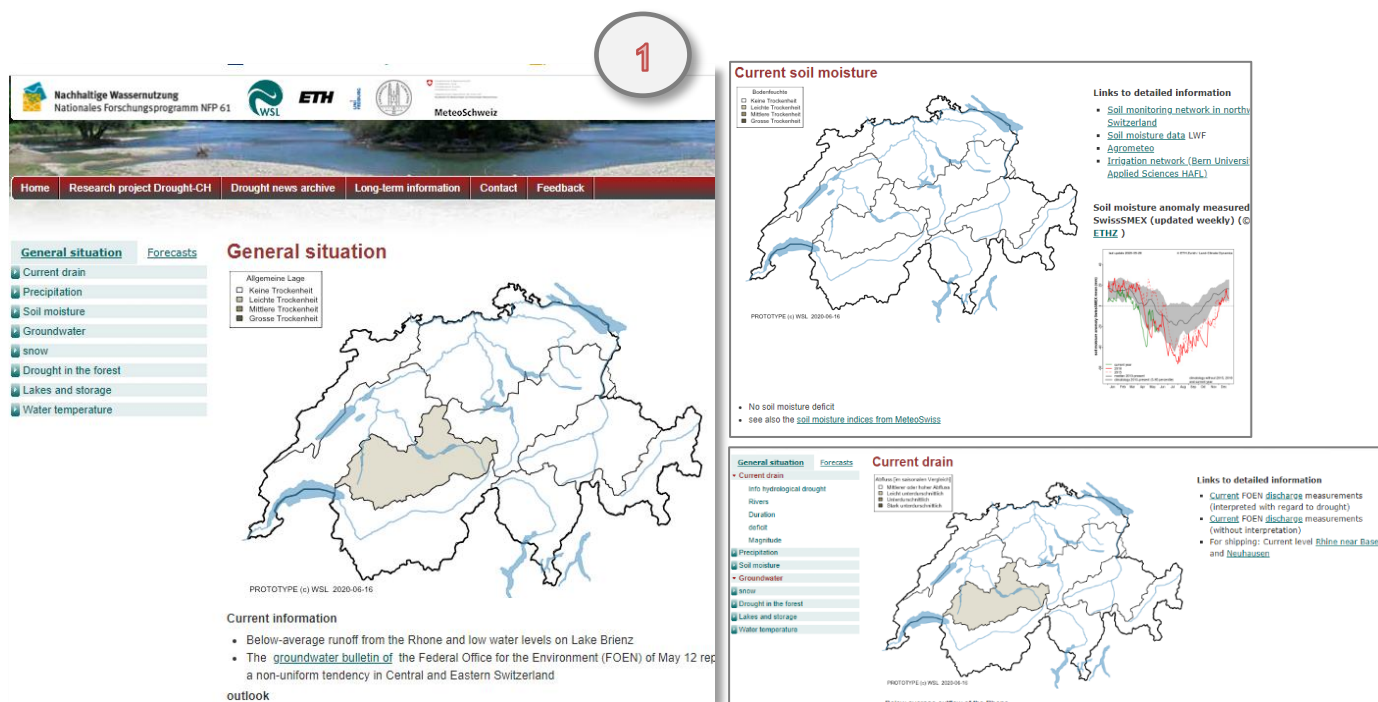
Bilje - kumulativna količina padavin

Ljubljana - kumulativna količina padavin

Lesce - kumulativna količina padavin

• EWS approach in Switzerland

Level of formality	Display of drought severity >> Communication to public	Link:
On the national level there is a platform for early detection of drought in Switzerland. Provides information on current and upcoming drought. These information are not official warnings.	<i>Country level:</i> Maps on current status on drought - precipitation, soil moisture, snow water equivalent, surface waters, groundwater, lakes, drought in the forest (classified in severity levels), water temperature; predictions; link to groundwater bulletin, graphs; articles	1. Platform for early detection of drought in Switzerland http://www.trockenheit.ch/ Prepared by WSL (developed in cooperation with MeteoSwiss and the FOEN; has been continued by the WSL)
	<i>Country level:</i> Maps – daily information related to drought indices, links to graphs (not classified in severity levels)	2. https://www.meteoswiss.admin.ch/home/climate/swiss-climate-in-detail/climate-indicators/drought-indices.html?param=rainanomaly Prepared by MeteoSwiss
	<i>Canton of Thurgau:</i> Maps - maps related to hydrology	3. https://map.geo.tg.ch Prepared by Canton of Thurgau



4. EXISTING DROUGHT PLATFORMS, REQUIREMENTS FOR ADO

Drought monitoring good practices

There are many requirements and examples of good practices of drought monitoring tools and early warning systems. Very generally, we can group them into two categories, which – at first glance – contradict each other:

- ease of use and clear information forwarded to the user on one hand and
- data/information richness on the other hand.

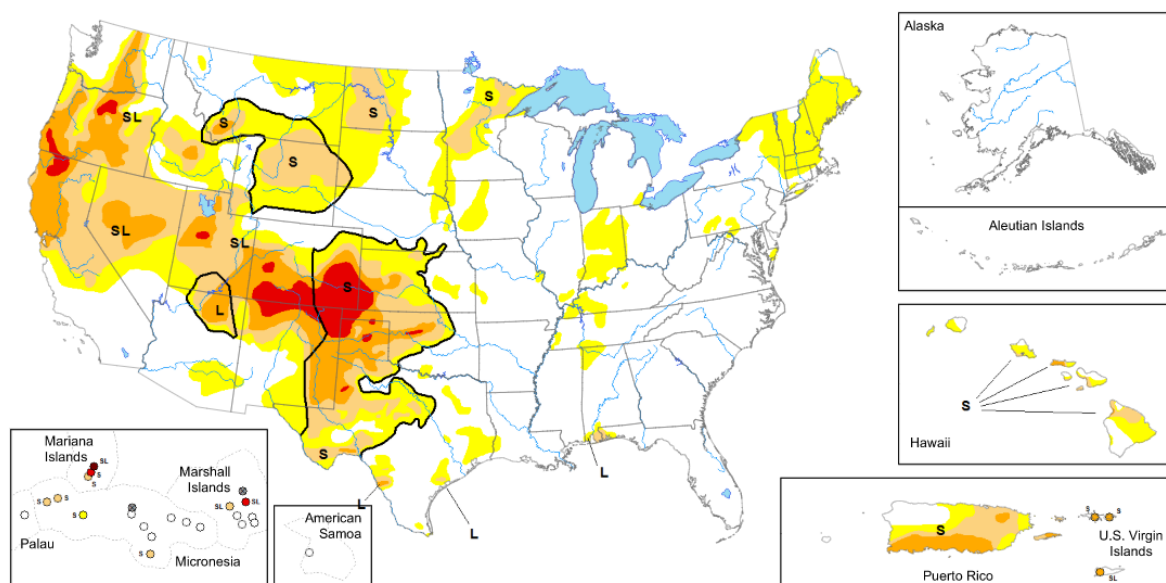
Ideal drought monitoring platform would overcome this antagonism and try offer best properties of each of two classes: usability for non-professionals and all information interested user might potentially need.

4.1 United States Drought Monitor (USDM)

USDM is usually used as reference for drought monitoring tools since it is operational over 20 years and was one of first web-based drought monitoring tools, intended for decision makers and general public. It belongs to the 1st group of DM tools with emphasis on ease of use. An example of overall map is presented in Figure 5.

Map released: June 18, 2020

Data valid: June 16, 2020



United States and Puerto Rico Author(s):
 Richard Tinker, NOAA/NWS/NCEP/CPC

U.S. Affiliated Pacific Islands and Virgin Islands Author(s):
 Curtis Riganti, National Drought Mitigation Center

The data cutoff for Drought Monitor maps is each Tuesday at 8 a.m. EDT. The maps, which are based on analysis of the data, are released each Thursday at 8:30 a.m. Eastern Time.

Intensity and Impacts

- | | |
|-----------------------|--------------------------|
| None | D3 (Extreme Drought) |
| D0 (Abnormally Dry) | D4 (Exceptional Drought) |
| D1 (Moderate Drought) | No Data |
| D2 (Severe Drought) | |

~ - Delineates dominant impacts

S - Short-Term impacts, typically less than 6 months (e.g. agriculture, grasslands)

L - Long-Term impacts, typically greater than 6 months (e.g. hydrology, ecology)

Figure 5: Example of US Drought Monitor map (available on <https://droughtmonitor.unl.edu>).

USDm has developed its own drought severity scale, marked (with clear reference to the hurricane Saffir-Simpson categories) as “D” categories from 0 to 4 (scale is included in the bottom of Figure 5). Although processing is automatic and “D” categories are determined objectively from blend of drought indicators (including PDSI, modelled soil moisture percentiles, SPI and streamflow percentiles; exact table is available on <https://droughtmonitor.unl.edu/About/AbouttheData/DroughtClassification.aspx>), final map is still finalized by hand and (as can be seen in Fig. 5) also signed by the author. Another two most notable properties of USDm are:

- separately from drought severity indicator there are areas of short-term impacts (mainly agriculture) and long-term impacts (hydrology, ecology) delineated in the map. Sources are numerous voluntary impact reporters, who are in touch with map author during time of preparation;
- although USDm now enables user certain “zoom” capabilities to individual states and even counties (which roughly correspond to NUTS3 regions), there is no web-GIS functionalities (such as underlying data with basemaps etc.) and no possibilities of obtaining data other than processed drought severity classes.

4.2 European Drought Observatory (EDO)

EDO is maintained by EC’s Joint research Center. Its main product, the map of Combined Drought Indicator (CDI) is presented in Figure 6.

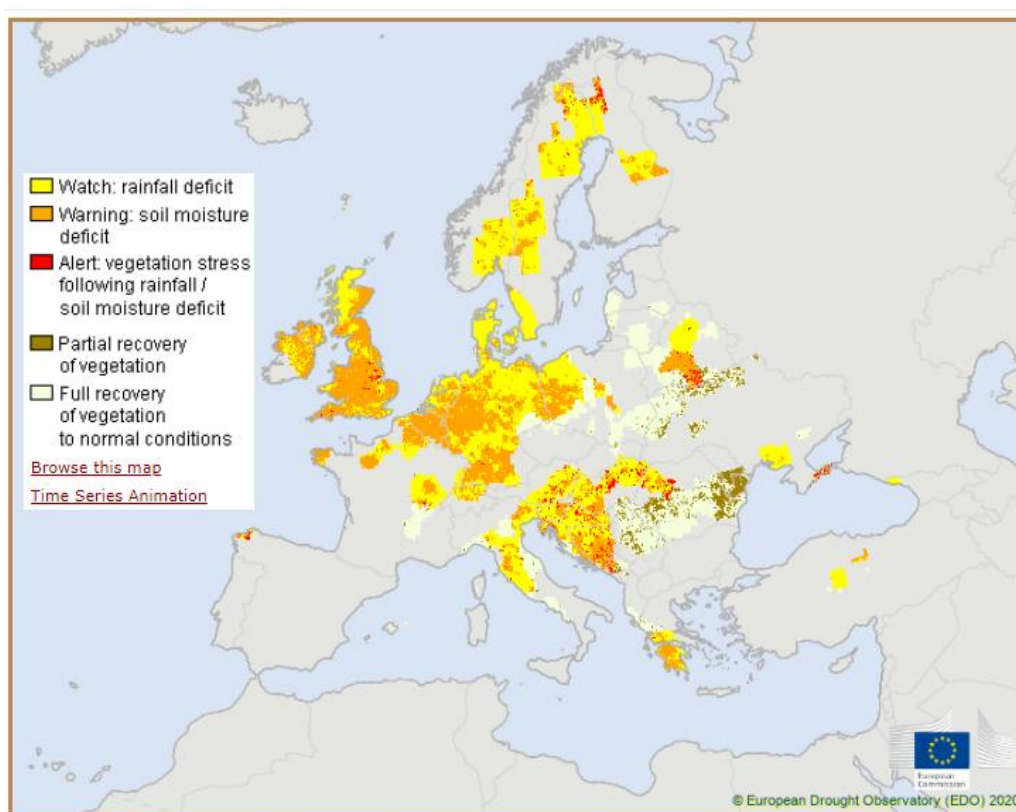


Figure 6: Example of EDO map (first third of June 2020; available on <https://edo.jrc.ec.europa.eu>)

CDI map is available as static picture in EDO home page, however by clicking on “Browse this map” link the user is directed to web-GIS map. Similar to USDm drought severity scale, also CDI is blend of various drought indicators (exact procedure is described here:

https://edo.jrc.ec.europa.eu/documents/factsheets/factsheet_combinedDroughtIndicator.pdf).

Apart from CDI values, user can browse also among other drought indices and prepare his own blend of drought indices (option “Make your own map”); as result of EuroGEOSS project introducing spatial data interoperability, there is also option to include maps of regional and local sources from their servers on main map.

We can therefore estimate that EDO offers both targeted properties of drought monitoring systems, however (since main CDI map is not general synthesis but just presentation of indicator blend) it offers less authoritative answer to question how severe is drought over specific region.

4.3 DroughtWatch.eu

Drought Watch tool is an open interactive web application that allows a harmonised view on changing drought conditions across the entire Danube region, and wider. It provides a spatial and temporal view of the state of soil moisture and vegetation through various drought-related datasets: remote sensing products, modelled indices and static products at regional and national level. Drought indices in the tool are updated daily, weekly or every 10 days, and available from 2007 onward, or earlier. This way, they enable a cross-border unified view of changing drought conditions on a fine scale and in near-real-time across the entire region. Drought Watch also includes a user manual with additional information on diverse functionalities of the tool and on datasets background. The tool complements regular drought monitoring through indices with integrated climatological and yield loss information on drought risk areas, and weekly information on drought impacts detected on the field, all in one place.

It is designed as a user-friendly yet feature-rich tool to strengthen the operational work in drought early warning. However, it does not provide simplified, single-scale look on situation (similar to USDM). It is therefore mainly designed for national authorities and drought experts. It is available also to other end-users such as farmers or water managers, so they can make appropriate decisions that lead to the application of relevant short-term measures. National authorities and expert services can further benefit from the possibility to integrate other national datasets into the Drought Watch tool.

Accessible at <https://droughtwatch.eu/>

- Video animation on how it works:
<https://www.youtube.com/watch?v=85tsBI6-8yM>
- Drought Watch Tutorial:
<https://www.youtube.com/watch?v=2MC5goO17H8>
- Drought Watch Manual:
www.droughtwatch.eu/#/manual

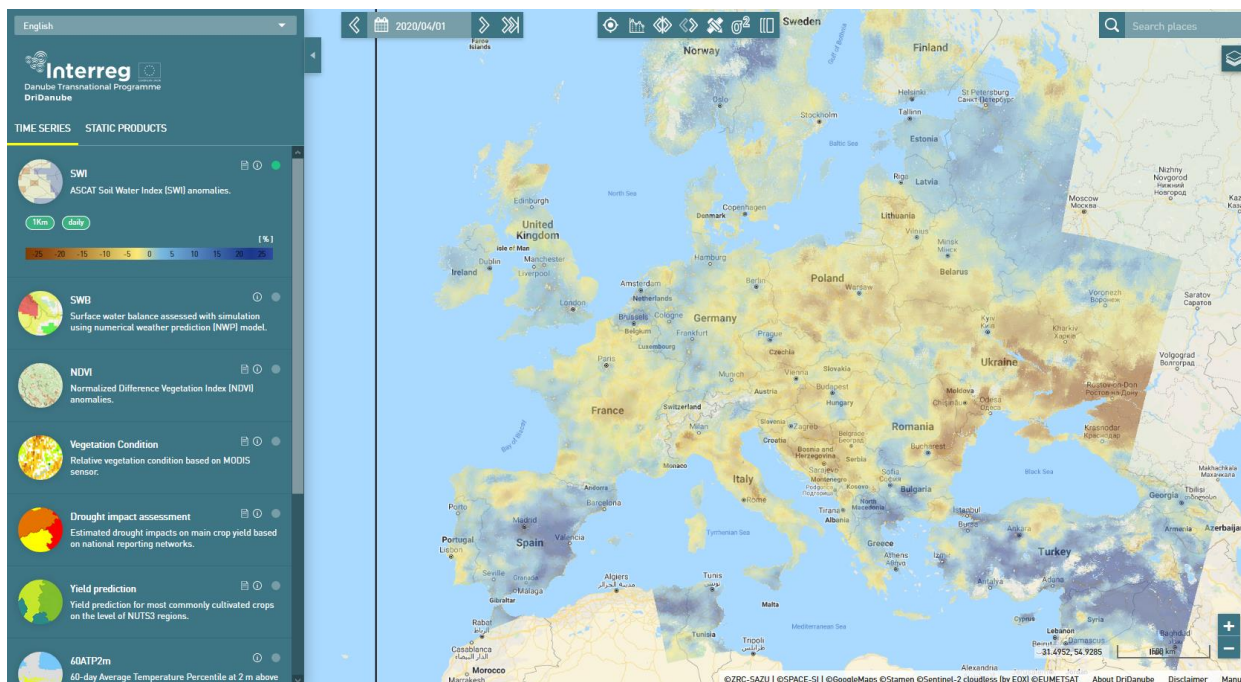


Figure 7: Drought Watch homepage, Soil Water Index for 1 April 2020.

4.4 Drought.ch

The aim of drought.ch (since the beginning of 1 June 2013) is to derive up-to-date information on drought from existing national monitoring networks and to present it in a way that is suitable for customers. The basic features and principles of the information platform are outlined:

- it should basically be a public information platform (not only for experts) and address all water use sectors (not specifically only for agriculture),
- the information should be presented on the spatial scale of regions,
- the intensity or the extent of a hydrological or meteorological parameter with regard to drought is classified and represented accordingly on the basis of a 30-year climatology. This means that the platform shows how rarely a corresponding drought-related situation is. However, it does not make any statement about the effects of the current drought,
- the platform is research-based, but should be easy to understand for everyone.

The structure of drought.ch shows a map whether there are current indications of drought anywhere in Switzerland. In addition, the site also provides current news information on drought. The platform is divided basically in the areas of a) information on the current situation section (runoff, precipitation, soil moisture, groundwater, snow, drought in forests, lakes and water temperature), and b) the forecasts section which shows ensemble forecasts of runoff, precipitation, soil moisture, groundwater, and snow water deficit.

The typified maps of the general situation are composed of the hydrometeorological parameters that are currently relevant to drought. As an example, Fig 8a shows the situation at the end of March 2015. If two or more hydrometeorological parameters provide indications of drought (in this example precipitation and the snow water equivalent), this is reflected in the general situation map. The size or its intensity determines the degree of colouring. Fig 8b provides a view of the climatological classification of the precipitation totals of the last 20 day in comparison with the values achieved in the past. Values less than 50% (shades of brown) indicate relatively dry conditions, values

above 50% (shades of green and blue) indicate rather wet conditions. The more intense is the colour (increase in chromaticity), the more unusual is the event.

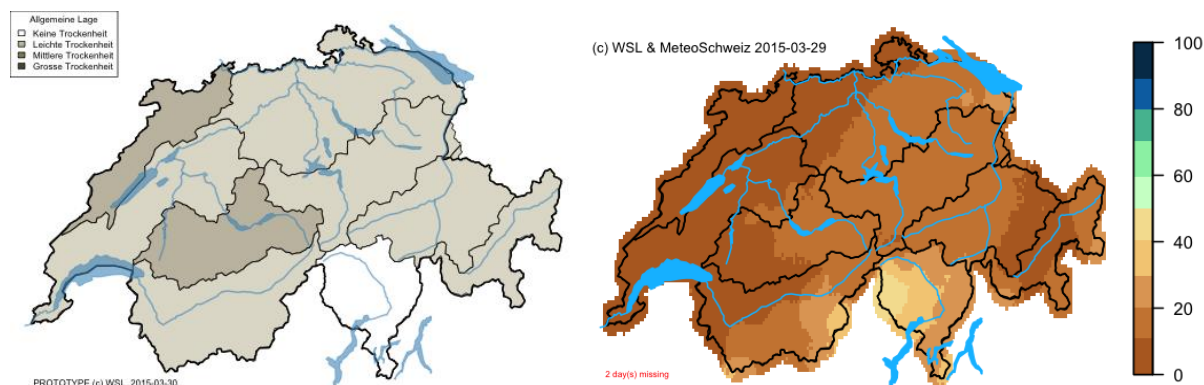


Figure 8: a) General map of 30 March 2015 on the entry page of drought.ch. b) Pixel map of the climatological classification of precipitation of the preceding 20 days before 29 March 2015 compared to the long-term average.

4.5 Conclusions

There are many approaches to construction of drought monitoring tools available on web; easily accessible data sources and web-GIS technology enabled various institutions to prepare their version of most appropriate monitoring tool. There is also no simple answer to question, whether it is better to include more information and let user decide which data to use, or is there an additional value in filtering and aggregating available data into simple, “traffic-light” style indicators. The latter case makes application of monitoring tool and decision-making process less complex, however important details might be lost in the process of preparation of simplified fields.

One of the most important question is also possibility to include drought impact data. If link to impacts is clear and collaboration with stakeholders and end-users is intense and continuous, it is easier to demonstrate clear link between monitoring and impacts and also to prepare simplified monitoring tool. If, on the other hand, the user is expected to interpret connection to impacts on his own, then it is better to make available wider set of indicators and data.

5. Annex

Collection of existing and proposed drought-related indicators and indices for potential integration into ADO platform.pdf